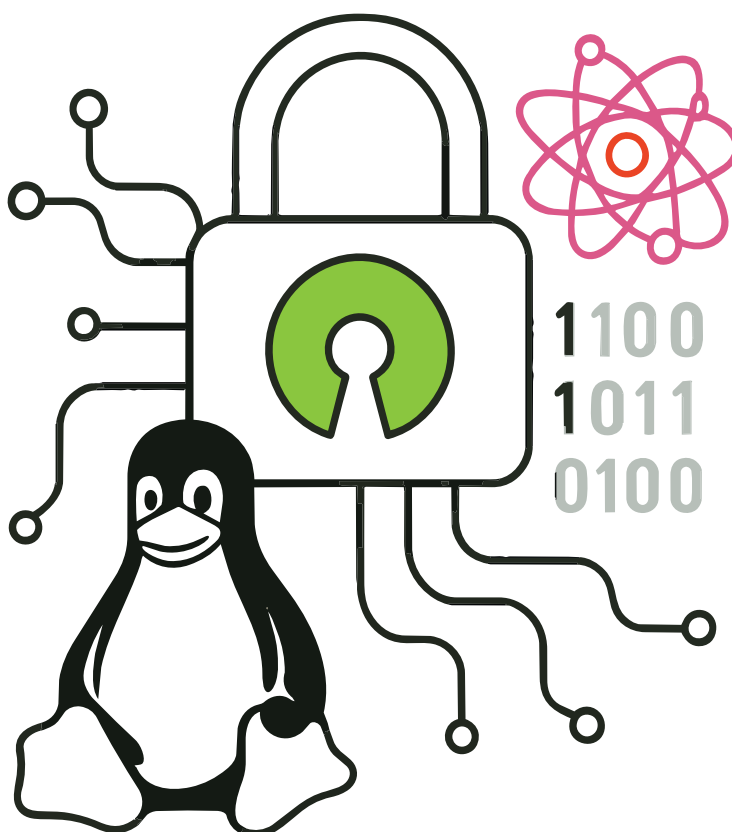


A Practical Guide to Securing Open Source in the Quantum Era

Quantum computing threatens today's cryptography. Open source must lead the migration to post-quantum cryptography (PQC) — and do it without breaking the ecosystem.



Quantum computing promises breakthroughs in medicine, materials, and AI. But the same machines that accelerate discovery will also shatter the encryption protecting the internet. Algorithms like RSA, ECDSA, and Diffie-Hellman — the backbone of TLS, VPNs, and secure messaging — will fall quickly to Shor's algorithm once large, error-corrected quantum computers arrive.

The timeline is uncertain: some predict a decade, others longer. But the threat is already here. Attackers can 'harvest now, decrypt later' — storing

encrypted traffic today and waiting until quantum machines can unlock it. That means yesterday's TLS sessions, VPN tunnels, and encrypted emails could be exposed tomorrow.

Open source sits at the centre of this risk. It powers the servers, registries, operating systems, and libraries the world depends on. Proprietary vendors can push controlled upgrades; open source projects are decentralised and slower to migrate. If upstream cryptographic libraries don't evolve, every downstream user remains vulnerable.

What is post-quantum cryptography?

Post-quantum cryptography (PQC) is classical software designed to resist quantum attacks. Unlike Quantum Key Distribution (QKD), PQC runs on ordinary hardware and can be deployed today.

The main families include:

- **Lattice-based schemes:** Fast, compact, and built on learning with errors (LWE).
- **Hash-based signatures:** Conservative, relying only on hash function security.